THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 5 & 6: **Parallel and Intersecting Lines, Number Play** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 43°	angles sum to 90°
2	**(b)** 145°	angles sum to 180°
3	**(b)** 75°	angles sum to 180°
4	**(d)** 125°	vertically opposite angles are equal
5	**(c)** 95°	linear pair sums to 180°
6	**(b)** 55	each term is the sum of the two before it
7	**(a)** odd	odd + even
8	**(b)** even	a product is odd only if both numbers are odd
9	**(b)** odd	parity rule
10	**(d)** Hardy–Ramanujan number	smallest sum of two cubes in two ways

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Four (2).	2
12	90° (2).	2
13	A = 8 (2).	2
14	9,801 vs 9,800; first is bigger (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	90 − x = x + 20 (1.5); x = 35° (1.5).	1½+1½
16	1, 3, 6, 10, 15, 21; differences 2, 3, 4, … (3).	3
17	180° straight (1.5); 120° obtuse (1.5).	1½+1½
	OR: 9x = 180 (1.5); 80° and 100° (1.5).	1½+1½
18	321 − 123 = 198; 198 + 891 = 1089 (3).	3
	OR: Only one odd number so sum is odd (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 105° (1). (ii) 75° (1). (iii) Two (1). (iv) Yes (2).	1+1+1+2
	OR: (i) 18x = 360, x = 20° (3). (ii) 100° (1). (iii) Yes (1).	3+1+1
20	(i) 11(a + b) = 121, so a + b = 11 (2). (ii) 56 and 47 (2). (iii) 10a + b + 10b + a = 11(a + b) (1).	2+2+1
	OR: (i) On diagonals going down-left (1). (ii) Each sums to 9 or 18 (1). (iii) All rows have one (1). (iv) Differences 9 (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 5 & 6: **Parallel and Intersecting Lines, Number Play** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 130°	corresponding angles are equal
2	**(a)** 95°	alternate angles are equal
3	**(d)** transversal	definition
4	**(d)** 180°	straight angle
5	**(c)** ruler	slide the set square along a ruler
6	**(b)** 4,512	4+5+1+2 = 12
7	**(d)** 1 to 9	each row, column, diagonal adds to 15
8	**(c)** square	10 × 10
9	**(d)** 18	3+4+5+6
10	**(d)** 25	squares

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	3x = 180°, x = 60° (2).	2
12	2x + 20 = 90; x = 35° (2).	2
13	Difference of alternate digit sums is 0 or multiple of 11: 1 + 1 − 2 = 0 (2).	2
14	1,527; 5,724 (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	72° and 108° (3).	3
16	Odd, as an odd count of odd numbers sums to odd (3).	3
17	Use a set square corner or measure 90° with a protractor (3).	3
	OR: 2x + 10 = x + 40 (1.5); x = 30 (1.5).	1½+1½
18	3 + 5 = 8 (3).	3
	OR: $8^2 = 64 > 60$ (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 5y + 30 = 180, y = 30 (2). (ii) 105° and 75° (2). (iii) 105°, 75° (1).	2+2+1
	OR: (i) Corresponding equal and vertically opposite equal (2). (ii) combine two properties (2). (iii) Ladder rungs, road design (1).	2+2+1
20	(i) 4n + 6 leaves remainder 2 (3). (ii) 18 not divisible by 4 (2).	3+2
	OR: (i) 45 (2). (ii) 9 × 10 ÷ 2 (2). (iii) 55 (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

Chapters 5 & 6: **Parallel and Intersecting Lines, Number Play** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 28°	angles sum to 90°
2	**(b)** 108°	angles sum to 180°
3	**(a)** 116°	angles sum to 180°
4	**(c)** 68°	vertically opposite angles are equal
5	**(b)** 47°	linear pair sums to 180°
6	**(b)** 89	each term is the sum of the two before it
7	**(b)** odd	even + odd
8	**(b)** even	a product is odd only if both numbers are odd
9	**(b)** even	has a factor 2
10	**(b)** 9	digit-sum rule

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Pairs of angles on opposite sides of the transversal, between the lines (2).	2
12	70° (2).	2
13	0 = 2 × 0 so it is divisible by 2 (2).	2
14	48, 96 (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	x = 75° (3).	3
16	K = 9; 81 (3).	3
17	Distance stays the same everywhere (3).	3
	OR: Alternate 52° (1.5); co-interior 128° (1.5).	1½+1½
18	4 + 7 + 8 = 19; 1 + 9 = 10; 1; not divisible (3).	3
	OR: 55 and 66 (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 120°; 15° (2). (ii) 75° (1). (iii) minute hand 180°, hour hand 195° (2).	2+1+2
	OR: (i) 58° (1). (ii) 58° (1). (iii) 122° (1). (iv) equal corresponding angles (2).	1+1+1+2
20	(i) 101, 111, 121 (2). (ii) 68 + 86 = 154; 154 + 451 = 605; 605 + 506 = 1111 (2). (iii) Repeated steps give palindrome (1).	2+2+1
	OR: (i) 24 (1). (ii) 210 (1). (iii) One is even and one multiple of 3 (2). (iv) No: 3 × 4 × 5 = 60 yes, 1 × 2 × 3 = 6 no (1).	1+1+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET D

Chapters 5 & 6: **Parallel and Intersecting Lines, Number Play** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 104°	corresponding angles are equal
2	**(c)** 41°	alternate angles are equal
3	**(b)** 180°	supplementary
4	**(b)** 360°	full turn
5	**(d)** parallel	converse of alternate angle property
6	**(d)** 25	$1+3+5+7+9 = 5^2$
7	**(a)** 5	middle of the arithmetic sequence
8	**(c)** odd	$n + (n + 1) = 2n + 1$
9	**(b)** 101	1 0 1
10	**(a)** 5	1, 3, 5, 7, 9

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Railway tracks (1); the corner of a table (1).	1+1
12	50° (2).	2
13	18 and 27 (2).	2
14	$36 = 6^2$ (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	270° (3).	3
16	Odd; ends in 5 (3).	3
17	$4x + 60 = 180$ (1.5); $x = 30°$ (1.5).	1½+1½
	OR: Opposite angles equal, adjacent are linear pairs (3).	3
18	1 + 3 + 5 + 7 + 9 + 11 (3).	3
	OR: $(10a + b) - (10b + a) = 9(a - b)$ (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) If corresponding angles are equal the lines are parallel (2). (ii) 80 = 80 (1). (iii) Not equal, so not parallel (2).	2+1+2
	OR: (i) 30° (2). (ii) 90° (2). (iii) No, right angle (1).	2+2+1
20	(i) $3 \times (100 + 10D + 5)$ digit sum: $1 + D + 5$ must be a multiple of 3 (2). (ii) D = 0, 3, 6, 9 give multiples of 9 after ×3 (2). (iii) $3 \times 105 = 315$, digit sum 9 (1).	2+2+1
	OR: (i) 210 (3). (ii) 21 × 10 pairs (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.